

100 Most Likely Defense Questions

Thesis: Wine Production Forecasting Using Rule-Based Machine Learning — A Distributional Subgroup Discovery Approach **Author:** Hugo Filipe Queiróz Nogueira · MECD, FEUP **Format:** Anticipated examination-committee questions, by topic. Each entry: *why the jury asks it* · *what knowledge it tests*. **Unanswered by design** — use as a drill sheet.

Note: Some questions (e.g. Q58) say "beam search" because the jury reads the submitted thesis, which used that wording. Per the errata, CAREN is an Apriori-style level-wise search — answer with the correct description (see

`01_defense_prep_azevedo.md`, Part 4).

Introduction

1. In one sentence, what is the scientific contribution of this thesis?

- *Why:* Tests whether you can separate contribution from activity.
- *Tests:* Ability to articulate a thesis-level claim, not a summary.

2. Is this thesis primarily a methods contribution or an application contribution?

- *Why:* Positions the whole defense; forces you to commit.
- *Tests:* Self-awareness of where the novelty actually lives.

3. Your title says "forecasting," yet forecasting fails. Is the title honest?

- *Why:* Probes for over-claiming in framing.
- *Tests:* Intellectual honesty; ability to reframe a null result.

4. Who is the intended beneficiary — data scientists or viticulturalists?

- *Why:* Interpretability is justified by the audience; they check it holds.
- *Tests:* Clarity of purpose and stakeholder reasoning.

5. Why two regions rather than one deeply studied region?

- *Why*: The two-region design is your strongest choice; they check you know why.
 - *Tests*: Understanding of controlled comparison vs. case study.
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Motivation

1. Why do we need interpretable rules if black-box models forecast better?

- *Why*: The core justification of the whole approach.
- *Tests*: Accuracy–interpretability trade-off reasoning.

2. You never quantify the accuracy cost of interpretability. How do you know it's worth paying?

- *Why*: You claim interpretability matters but never measured the ceiling.
- *Tests*: Awareness of the missing black-box benchmark.

3. Is annual regional production actually a decision-relevant quantity for producers?

- *Why*: Challenges the practical motivation.
- *Tests*: Domain grounding of the problem statement.

4. SHAP gives post-hoc explanations. Why is that insufficient for your use case?

- *Why*: Tests the distinction you draw between auditable and approximate explanations.
- *Tests*: Understanding of explainability methods.

5. What real decision would change if a producer had your rules?

- *Why*: Forces a concrete impact statement.
- *Tests*: Translation of statistical output into action.

6. Why is this an MSc-worthy problem rather than an engineering exercise?

- *Why*: Establishes scientific weight.
 - *Tests*: Ability to defend the research question's depth.
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Wine production

1. Why model total production instead of yield per hectare?

- *Why*: The most dangerous domain question; area is a structural confound.
- *Tests*: Whether you understand $\text{production} = \text{area} \times \text{yield/ha}$.

2. The benefício quota administratively caps Port output. How does that affect your target?

- *Why*: An unobserved covariate decouples production from climate.
- *Tests*: Awareness of non-climate drivers you cannot see.

3. Explain the two-year vine cycle and why it justifies your y0 features.

- *Why*: Core agronomy behind your feature engineering.
- *Tests*: Grapevine physiology (primordia, reserves).

4. What agronomic mechanism makes moderate soil-water deficit *beneficial* in the Douro?

- *Why*: Your top RDD rule depends on it.
- *Tests*: Berry-concentration vs. dilution physiology.

5. Why does the same soil-water feature invert direction between regions?

- *Why*: Your headline method-validation point.
- *Tests*: Understanding of climate context dependence.

6. How much of the Vinho Verde decline is climate vs. EU policy?

- *Why*: The RVV confound is central to your failure analysis.
- *Tests*: Grasp of the structural drivers (grubbing-up, pergola→espalier).

7. Are STICS phenological dates and VSIM soil water measurements or model outputs?

- *Why*: Two of your key variables are modelled, not observed.
 - *Tests*: Honesty about data provenance and error propagation.
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Climate

1. Which climate variables dominate Douro production, and per whose prior work?

- *Why*: Grounds your features in the literature (Cunha).
- *Tests*: Knowledge of spring temperature / soil water drivers.

2. How do you separate a warming trend from within-era climate signal?

- *Why*: This is the era-confound problem in miniature.
- *Tests*: Understanding of the LOESS retention diagnostic.

3. Harvest temperature shows era inflation. Why is that a red flag?

- *Why*: Only 56% LOESS retention — a co-trending artifact.
- *Tests*: Ability to distinguish real signal from smooth co-trend.

4. Why phenology-anchored windows instead of fixed calendar months?

- *Why*: Justifies the feature construction choice.
- *Tests*: Understanding that phenology shifts 2–4 weeks yearly.

5. Does climate change threaten the stationarity your rules assume?

- *Why*: Rules are backward-looking; future climate departs from history.
- *Tests*: Awareness of non-stationarity under warming.

6. Could a single extreme weather event dominate a year's residual and break a rule?

- *Why*: Regional aggregates hide within-season shocks.
- *Tests*: Understanding of unexplained variance sources.

Statistics

1. The $p = 0.031$ result is post-hoc with $n = 5$. Defend or retract it.

- *Why*: HARKing plus tiny sample.

- *Tests:* Confirmatory vs. exploratory discipline.

2. You applied no multiple-comparison correction across 17 models. Justify.

- *Why:* Family-wise error inflation.
- *Tests:* Multiplicity awareness.

3. Your 18 walk-forward scenarios share most of their test years. Is the Wilcoxon test valid?

- *Why:* Overlapping test sets violate independence.
- *Tests:* Understanding of test assumptions and autocorrelation.

4. You mine thousands of subgroups at KS $p \leq 0.10$. How many of your 48 rules are false positives?

- *Why:* Search-space multiplicity inside rule mining.
- *Tests:* Whether you understand the null-distribution problem of subgroup search.

5. Why KS $p \leq 0.10$ rather than 0.05?

- *Why:* A lenient threshold under heavy multiplicity.
- *Tests:* Justification of the significance level.

6. Explain the KS statistic and why it suits $n = 80$.

- *Why:* Foundational to CarenR.
- *Tests:* Non-parametric ECDF comparison knowledge.

7. Why MAE, when the median baseline is by construction the MAE-minimizer?

- *Why:* You chose the loss your baseline is optimal under.
- *Tests:* Metric-selection reasoning; would RMSE change the story?

8. What is the statistical power of a test on 5–18 scenarios?

- *Why:* Underpowered design.
- *Tests:* Power analysis literacy.

9. ADF and KPSS disagree for RVV residuals. What does that tell you?

- *Why*: Mixed stationarity verdict is diagnostic.
- *Tests*: Understanding of unit-root vs. trend-stationarity.

10. Why the Wilcoxon signed-rank rather than a paired t-test?

- *Why*: Distributional assumptions on small samples.
- *Tests*: Test-selection rationale.

Machine Learning

1. Why did you not include a penalized regression (LASSO/elastic net) baseline?

- *Why*: The obvious interpretable competitor is missing.
- *Tests*: Breadth of ML baseline knowledge.

2. Where is a classical time-series model (ARIMA/ARIMAX)?

- *Why*: A forecasting thesis without ARIMA is conspicuous.
- *Tests*: Time-series forecasting fundamentals.

3. What is the predictive ceiling? What would Random Forest or gradient boosting achieve on *your* data?

- *Why*: You cite RF from others but never ran it.
- *Tests*: Willingness to establish an upper bound.

4. How do you prevent data leakage in the walk-forward pipeline?

- *Why*: The credibility of every predictive number depends on it.
- *Tests*: Understanding of within-fold preprocessing.

5. Your feature/observation ratio is 59:80. Isn't that a curse-of-dimensionality problem?

- *Why*: Overfitting risk is severe.
- *Tests*: Dimensionality and sample-size reasoning.

6. How do you handle collinearity among near-duplicate features?

- *Why*: Combinatorial features are highly correlated.
- *Tests*: Feature-space hygiene.

7. Is feature selection stable across folds? Do the same features get chosen each time?

- *Why*: Unstable selection undermines interpretability.
- *Tests*: Whether you measured selection stability (you did not report it).

8. Decision Tree performs worst. Does that mean tree structure is wrong for this data, or the tree was mis-tuned?

- *Why*: Distinguishes model failure from configuration failure.
- *Tests*: Understanding of hyperparameter effects.

Rule Learning

1. Define subgroup discovery and how it differs from classification.

- *Why*: Foundational vocabulary.
- *Tests*: Descriptive vs. predictive induction.

2. Why are rule-based methods "interpretable" in a way SHAP is not?

- *Why*: The central justification of the family.
- *Tests*: Explicit-rule vs. attribution reasoning.

3. What is the difference between association rules and distribution rules?

- *Why*: CarenR is built on distribution rules.
- *Tests*: Understanding of preserving the full target distribution.

4. Rules overlap 65% pairwise. Isn't your rule set redundant?

- *Why*: Challenges the coherence claim.
- *Tests*: Understanding of overlapping-subgroup design.

5. How do you set the minimum support threshold, and why 20%?

- *Why*: An experimental design choice, not algorithmic.

- *Tests*: Support/robustness trade-off reasoning.

6. Why are older learners (RIPPER 1995) your comparators rather than modern ones (RuleFit, CORELS, Bayesian Rule Lists)?

- *Why*: Your comparator set skews old.
- *Tests*: Awareness of the current interpretable-ML landscape.

7. How do you convert descriptive rules into a point forecast?

- *Why*: The discovery→prediction bridge.
- *Tests*: Understanding of the aggregation mechanism.

CarenR

1. Walk me through the CarenR pipeline end to end.

- *Why*: Establishes command of your primary method.
- *Tests*: Discretization → level-wise (Apriori) search → KS → support → aggregation.

2. Which parts of Figure 3.3 are the CarenR algorithm and which are your thesis design choices?

- *Why*: You must own what you added vs. inherited.
- *Tests*: Boundary between framework and contribution.

3. Why equal-frequency quartile binning rather than equal-width or supervised bins?

- *Why*: Discretization drives rule conditions.
- *Tests*: KS-reliability rationale for balanced bins.

4. CarenR_Supervised was your worst variant. Why does supervised discretization fail?

- *Why*: A clean negative result about target leakage.
- *Tests*: Understanding of leakage in feature engineering.

5. Support-weighted median aggregation is ad hoc. Why not a learned

combiner?

- *Why*: The prediction rule is heuristic.
- *Tests*: Justification of the aggregation formula.

6. Why does η^2 win in RDD but Sup_FS win in RVV?

- *Why*: The rank reversal is a key result.
- *Tests*: Understanding of signal-strength vs. selection-conservatism.

7. The fallback makes CarenR default to the median. Doesn't that guarantee it can't beat the baseline?

- *Why*: Probes the ceiling imposed by your own design.
- *Tests*: Understanding that value can only come from fired rules.

8. Is a model that abstains 44% of the time and harms 56% of the time in RVV actually useful?

- *Why*: Challenges the fallback-as-virtue framing.
- *Tests*: Honest appraisal of net utility.

9. Beam search is greedy. Could it miss the best subgroups?

- *Why*: Search completeness. (*Note: the premise is wrong — CAREN is level-wise/exhaustive over the support-frequent region. Correct it: it does NOT miss frequent subgroups the way a beam would.*)
- *Tests*: Understanding of heuristic vs. exhaustive search.

10. Prior CarenR work was in-sample only. What exactly is new in your use of it?

- *Why*: Pins down your incremental contribution.
- *Tests*: Positioning against Moreira 2022 / Jorge 2006.

RIPPER

1. Why is RIPPER a fair comparator when it needs a discretized target?

- *Why*: The tertile discretization loses information.
- *Tests*: Fairness of cross-method comparison.

2. Explain FOIL gain and MDL pruning.

- *Why*: Core RIPPER mechanics.
- *Tests*: Rule-growing and overfitting control.

3. RIPPER's default rule covers most years. What does that failure mode tell you?

- *Why*: The default rule dominates on small data.
- *Tests*: Understanding of class-imbalance and coverage.

4. Cross-validated accuracy is ~28–42%. Is RIPPER contributing anything?

- *Why*: Barely above the 33% random baseline.
- *Tests*: When a comparator is informative despite poor accuracy.

5. Your "CV accuracy" — is that random k-fold? Doesn't that violate your own no-random-CV stance?

- *Why*: Potential internal inconsistency with Ch.2.
- *Tests*: Consistency of methodology.

M5Rules

1. M5Rules collapsed to a global linear model. Why include it at all?

- *Why*: It never split — is it a real comparator?
- *Tests*: Understanding of why splits fail at $n \approx 80$, leaf-min 4.

2. Explain standard-deviation-reduction splitting and leaf smoothing.

- *Why*: Core M5 mechanics.
- *Tests*: Model-tree construction knowledge.

3. M5Rules gives a negative soil-water slope; CarenR gives a positive moderate-deficit range. Which is right?

- *Why*: An apparent contradiction between methods.
- *Tests*: Linear-marginal vs. non-linear bin-specific structure.

4. CV R^2 is negative for RVV. Why report coefficients from an overfit model at all?

- *Why*: Interpreting coefficients from a model worse than the mean.
- *Tests*: Discipline about what is interpretable.

5. Does M5Rules' collapse mean the relationship is truly linear, or that the method is starved of data?

- *Why*: Two very different conclusions.
- *Tests*: Distinguishing model limitation from data structure.

Evaluation

1. Justify the weighted-MAE metric. Why weight by test-set size?

- *Why*: Weighting interacts with era composition.
- *Tests*: Understanding of how the metric is constructed.

2. Weighting gives early, most-confounded scenarios the most influence. Isn't that backwards?

- *Why*: The metric penalizes CarenR precisely where it's weakest.
- *Tests*: Awareness of metric–design interaction.

3. Your protocol text says "predict the next year," but tables predict all remaining years. Which did you do?

- *Why*: An internal inconsistency in the design description.
- *Tests*: Precision about your own experiment.

4. Multi-step trend extrapolation dominates early scenarios. Is your MAE measuring climate or trend?

- *Why*: The residual re-trending step carries most of the error.
- *Tests*: Decomposition of error sources.

5. Only 17 test years per region across ~3 production cycles. Can any conclusion be robust?

- *Why*: The evaluation window is short.
- *Tests*: Honesty about statistical reach.

6. Why walk-forward instead of blocked/nested CV designs?

- *Why*: Alternative temporally-valid schemes exist.
- *Tests*: Knowledge of time-series validation options.

7. How would a permutation or block-bootstrap test change your significance claims?

- *Why*: A more honest null for dependent data.
- *Tests*: Resampling literacy.

8. What baseline would truly be hard to beat — and did you use it?

- *Why*: Median is easy under MAE; ARIMAX or persistence-with-drift is harder.
- *Tests*: Baseline rigor.

Prediction

1. State plainly: did any model beat the naive baseline? Yes or no.

- *Why*: Forces the null result into the open.
- *Tests*: Directness under pressure.

2. Why do the same rules that describe the past fail to predict the future?

- *Why*: The central asymmetry.
- *Tests*: Discovery vs. prediction reasoning.

3. In RVV, adding modern data makes CarenR worse. Explain the mechanism.

- *Why*: Counter-intuitive era-tuning.
- *Tests*: Understanding of era-specific overfitting.

4. If $Irl \approx 0.47$ in-sample, why doesn't that translate to out-of-sample skill?

- *Why*: In-sample correlation $\neq$ predictive value.
- *Tests*: Generalization gap reasoning.

5. **The era-balanced window wins 5/5, then loses 2019–2022. Doesn't that destroy the "promise"?**
 - *Why*: The promising window collapses immediately after.
 - *Tests*: Robustness of the conditional claim.
 6. **Is a directional (above/below trend) forecast more achievable than a point forecast? Did you test it?**
 - *Why*: A weaker but useful target you did not evaluate.
 - *Tests*: Reframing the prediction task.
 7. **How would a producer use a model that beats the median only under specific era conditions?**
 - *Why*: Practical deployability of a conditional result.
 - *Tests*: Bridging statistics to use.
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Discussion

1. **Is the two-decision decomposition genuinely novel, or a restatement of error decomposition?**
 - *Why*: Your best contribution challenged on originality.
 - *Tests*: Precise articulation of what is new.
2. **You say Decision 1 is the bottleneck. Prove it isn't Decision 2 (weak rules).**
 - *Why*: The diagnosis must be defended.
 - *Tests*: Evidence that the rule step is near the signal ceiling.
3. **Why do the two regions diverge so sharply — degree or kind?**
 - *Why*: Structural vs. gradational difference.
 - *Tests*: Understanding of confound magnitude.
4. **Cross-algorithm convergence — could it reflect shared confounding rather than truth?**
 - *Why*: Three methods on the same confounded features may agree wrongly.

- *Tests*: Skepticism about your own robustness argument.

5. Your $I_{rl} \approx 0.47$ matches Fraga et al. Does that mean the signal ceiling is intrinsic?

- *Why*: Positions your result against prior work.
- *Tests*: Literature-grounded interpretation.

6. If you re-ran everything on yield/ha, what would you predict changes?

- *Why*: Forces a falsifiable prediction about your own fix.
- *Tests*: Scientific reasoning about the confound.

Limitations

1. What is the single most serious limitation, in your own words?

- *Why*: Self-critique maturity.
- *Tests*: Prioritization of weaknesses.

2. Two of your key variables (soil water, LAI) are model outputs. How does that error propagate?

- *Why*: VSIM/STICS uncertainty is unquantified.
- *Tests*: Error-propagation awareness.

3. You never validate rules on truly held-out data for Goal 1. Is "validated" the right word?

- *Why*: Goal 1 validation is all in-sample.
- *Tests*: Precision of the "validated" claim.

4. The abstract says 90–92% LOESS retention; Table 4.4 shows 15–16% for RVV. Which is correct?

- *Why*: A direct internal contradiction.
- *Tests*: Command of your own numbers.

5. Single-region annual aggregates preclude spatial analysis. Does that limit the whole conclusion?

- *Why*: Aggregation hides heterogeneity.
- *Tests*: Understanding of resolution limits.

Future Work

1. You propose structural detrending as the fix. Why didn't you do it within the thesis?

- *Why*: The central prescription is left undone.
- *Tests*: Scoping justification and feasibility awareness.

2. What exactly would go into the structural covariate vector, and is the pre-1986 data available?

- *Why*: Feasibility of your own proposal.
- *Tests*: Concrete grasp of the data sources (IVV/IVDP).

3. Why would EMM detect interactions that CarenR's marginal KS rules miss?

- *Why*: You cite EMM as the natural generalization.
- *Tests*: Understanding of model-based interestingness.

4. How would you pre-register the era-balanced hypothesis for a confirmatory test?

- *Why*: Turns the post-hoc finding into science.
- *Tests*: Knowledge of pre-registration and prospective design.

5. If a new student continued this, what is the first experiment they should run?

- *Why*: Tests whether you know the highest-leverage next step.
- *Tests*: Research prioritization.